Climate Adaptation Resilience: An Assessment of SHARPE

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Introduction

This report summarises the findings of an assessment of the **FCDO funded Strengthening Host and Refugees Populations in Ethiopia (SHARPE)** programme on the contribution of its interventions in strengthening the climate resilience of refugee and host communities.

The purpose of the assessment was to consider the climate risks to SHARPE's interventions, what could be done to strengthen adaptive capacity, and how these adaptation measures could be integrated into its interventions. The assessment also considered SHARPE's contribution to the climate adaptation and mitigation of the refugee and host communities in which it works. Looking to the future, this report suggests actions in current and future programmes to strengthen the climate resilience of refugee and other vulnerable communities.

SHARPE works in refugee and host communities in two high-risk regions, Gambella and Somali. SHARPE aims to catalyse transformational system change to the protracted refugee contexts in Ethiopia by using a market systems development approach to strengthen the economies of the target communities in Dollo Ado and Jijiga (Somali region) and Gambella. SHARPE works to increase refugee self-reliance and generate economic opportunities through the piloting and scaling of private sector led interventions. SHARPE also aims to serve as a learning platform to share lessons in the application of market-oriented systemic approaches in humanitarian settings.

Climate Change within the Ethiopian Context

Ethiopia is one of most vulnerable countries in the world to climate change. According to the ND-GAIN index[1] (2023) for climate vulnerability, Ethiopia ranks 163 out of 185 countries, ranked the 24th most vulnerable country and the 156th most ready country meaning that it is vulnerable to, yet largely unready to combat climate change effects.[2] Although globally one of the most vulnerable countries, Ethiopia is a low greenhouse gas emitter, contributing only 0.36% of global GHG emissions (2022).[3]

With almost 90% of its surface vulnerable to severe or extreme climate stresses, climate change poses a huge challenge for Ethiopia. It is one of the world's most drought prone countries. Drought, often linked closely with El Niño events, has been and continues to be the most damaging climate stressor in the country. Ethiopia is experiencing increasingly unpredictable rains, and in some years the complete failure of seasonal rains, occurrences that are linked to climate change. Drought and high temperature extremes are the main drivers hindering crop and livestock productivity. The economy is dominated by agriculture which is primarily rainfed employing about 64 percent of the population, most of whom are smallholders. Ethiopia also has the largest livestock population in Africa. In the drier and hotter lowlands, livelihoods are generally based on raising livestock instead of farming.[4]

Since 1960 Ethiopia's climate trends include[5,6]

- Annual average **increase in temperature** by 1.7°C (1960-2020), with most noticeable increases in already dry and hot areas of the country and in the July-September season.
- An overall **decline in precipitation** levels in the last three to four decades, with regional variability and significant year-to-year volatility.
- Changes in temperature and rainfall have **increased the frequency and severity of extreme events**. Warming has exacerbated droughts, and desertification in the lowlands is expanding.

Climate projections are for the current trends to continue with rising temperatures, uncertain rainfall, more hot days, fewer cold days and nights and an increase in extreme events, especially droughts.

Figure 2: Climate projections

- **Projected increase in temperature of 1.8°C by 2025**
- **Erratic rainfall, increased unpredictability of seasonal rains, more frequent heavy precipitation events**
- **More frequent and severe dry spells and drought events**

Regional Variations

Reflecting topographical differences across the country, there are large regional differences to climate risk vulnerability. More than half of the country is higher than 1,500 meters above sea level, but there are also low-lying areas of tropical savannah and desert, particularly in border regions where the majority of refugees are housed. Precipitation and water availability is strongly dependent on elevation – highland areas exhibit mean annual temperatures of 11.9°C (Amhara) and more than 1,900 mm of rain, while the lowlands exhibit mean annual temperatures of up to 30.8°C and annual precipitation of only 100 mm in the lowlands. The highlands are projected to experience extreme events such as dry spells coupled with irregular and intense rainfall. The lowland regions are projected to be at greater climate risk with prolonged droughts and increasing temperatures affecting the agropastoral production system in the lowland regions. Water supply will be affected throughout the country, but most severely in areas that are already water-scarce, such as densely populated parts of the Rift Valley and eastern pastoralist zones.

Climate risk and vulnerability in refugee hosting regions – Somali and Gambella

Ethiopia hosts the second largest refugee population in Africa with over million refugees (1,051,000 in May 2024)[7] primarily from Somalia, South Sudan and Eritrea, many of whom have been in Ethiopia for more than 10 years and are unlikely to return home. While Ethiopia is now moving to a settlement and integration approach, historically the approach has been to house refugees in camps, most of which are located in border regions which are also some of the poorest and most underdeveloped regions of Ethiopia. These border regions are highly vulnerable to climate change, have limited income generating opportunities and competition over land and resources causes frequent conflict. The Gambella and Somali regions together host approximately 740,000 refugees, originally from South Sudan and Somalia.[8]

The Somali region has already become drier, hotter and more unpredictable over past years, and is expected to worsen in the future. The region is predicted to witness the highest increase in very hot days over the next decades and extreme precipitation events are expected to become more frequent and more intense. Although mean annual rainfall may remain stable or even increase over the long-term, the incidence of droughts and flooding is expected to increase throughout the century. With more heatwaves, higher rainfall variability and an increase in heavy precipitation events, the Somali region is and will remain in the future highly exposed to the extreme weather events not only of droughts but also extremely high risk from flooding.

The high dependence on pastoralism and rain-fed agriculture makes the Somali region highly vulnerable to various climate shocks, especially drought. Research conducted in the Somali region in December 2023[9] reported a sharp decline in the number of households reporting owning livestock since the beginning of the droughts in 2020, from 43% to only 18% due to the death of livestock. In 2023, 3.9 million people were estimated to need emergency food assistance and the number of people facing critical water shortage and requiring emergency water response was estimated at twenty-nine million (March 2023).[10]

The Somali region is facing a major threat to its grazing areas and livestock from the rapid expansion of a pernicious invasive species, Prosopis Juliflora which is well adapted to the drier and hotter conditions of climate change, withstands floods, grows well in degraded grasslands and is hard to effectively control and tehrefore is likely to transform the current landscape into an area that will support far fewer livestock.[11]

Gambella is one of the most vulnerable regions to climate change and is also one of the most marginalised and conflict-ridden regions in Ethiopia, with the large refugee population from South Sudan exacerbating tensions between the indigenous Anuak and Nuer ethnic groups. Low and humid, encompassing several river basins and the country's sole tropical mountain rainforest, the greatest climate risks in Gambella will be from **erratic rainfall and floods**, such as the recent floods in 2023 that are increasing in freqency and magnitude, and **rising temperatures** which will affect agriculture and livestock.[12] **The changing rainfall pattern** in combination with **warming trends** could make rainfed agriculture riskier and reduce food production in the region. Climate risks can exacerbate the inter-communal tensions and conflict in Gambella which in turn increase the vulnerability of both hosts and refugee communities to current and future climate risks.

Vulnerability of Refugees

Refugees experience bio-physical-economic risks associated with climate change in their areas and the socio-economic constraints of their refugee status. As seen for refugees globally, refugees in Ethiopia are poorer, more marginalised and politically disenfranchised compared to the host population. Mostrefugees live in camps and receive regular humanitarian assistance. Ethiopian policy is moving towards socio-economic inclusion of refugees into Ethiopian society, but operationalisation of these more inclusionary policies is lagging and is hindering the pace of the anticipated inclusion. Continued lack of basic socio-economic rights for refugees translates into less resilience to severe weather events that undermines their agriculture, pastoral livelihoods and food security. Building the adaptive capacity of refugee and host communities is crucial to withstand climate variability and change.

Strengthening Host and Refugee Populations in Ethiopia (SHARPE)

Using a market system development approach SHARPE promotes inclusive and sustainable economic development for refugee and host communities in the Gambella and Somali regions. SHARPE facilitates private sector investment and market linkages into these refugee hosting regions to generate inclusive economic opportunities for both refugees and host communities. SHARPE has catalysed investment in strategic engagement areas in agri-business, solar energy, and finance, offering co-investment, technical assistance and capacity development to partners on a cost-sharing basis.

SHARPE has generated successes with sustainable commercial businesses developed among both host and refugee communities in agri-business (egg and poultry production, livestock, feed, agro-vets and animal health services, agri-inputs, crop production) digital finance and solar energy. These micro-businesses operating inside and around the camps are linked with regional MSMEs to access a range of support services, including finance, technical support and essential inputs, creating market relationships and supply chains that are expected to be sustainable beyond the lifetime of the SHARPE programme.

Climate Assessment of SHARPE Interventions

This climate risk assessment of the SHARPE programme focuses on the climate risks associated with the regions in which it is working, the contribution of its current programme to climate resilience, adaptation, and mitigation, and recommendations on how SHARPE could further strengthen resilience and enable local adaptation to climate risks and mitigation (e.g. lower GHG emissions).

In this context adaptation refers to the capacity of communities and households to adjust their economic system in response to actual and/or expected climate changes/extremes in order to moderate potential damage and/or to benefit from opportunities associated with climate change. Adaptation activities provide the resources, tools or skills to better cope with changing, unpredictable and extreme weather. While adaptation is the process of adjusting to the actual or expected climate and its effects, resilience is the actual capacity to prepare for these effects, particularly the capacity to 'bounce back' from destructive climate-related events.[13]

Due to high vulnerability to weather shocks, livelihood diversification is commonly used as a strategy by agricultural, pastoral and agropastoral households for risk management. An important role that programmes such as SHARPE can play to facilitate climate adaptation and strengthen resilience is support livelihood diversification by resolving the financial bottlenecks of pastoral households and supporting the establishment and expansion of micro and small enterprises in semi-arid areas.[14, 15]

The climate assessment focused on the interventions that were implemented in Dollo Ado in the southern part of the Somali region, specificially, poultry and egg production; sheep and goat trading and fattening; agro-vet and animal health services; solar energy; and financial inclusion. Interventions on climate smart agriculture implemented in Gambella were not included in this assessment.

Home to 210,000 Somali refugees housed in five camps, Dollo Ado has been referred to by UNHCR as one of the 'epicentres of the climate crisis'. After five consecutive failed rainy seasons, the drought (2020-23) resulted in water shortages that destroyed thousands of acres of crops, killed hundreds of thousands of livestock, and dried up water sources in many communities. The drought was followed by dangerous flash floods and the destruction of agricultural fields, and infrastructure – water systems, irrigation, roads and bridges.

Poultry

Climate risk. Increasing temperature is likely the greatest climate risk factor affecting poultry. Poultry are particularly vulnerable to climate change because birds can only tolerate narrow temperature ranges with poultry production decreasing at temperatures higher than 30°C. Heat stress on poultry reduces feed intake, weight gain, carcass weight, and protein/muscle calorie content. Laying hens respond to increasing temperatures by reducing the number and size of eggs produced. Stressful weather conditions negatively affect internal and external quality of eggs. High temperatures result in lower fertility rates, egg hatchability, and chick quality. Additionally, heat-stressed males produce semen with lower sperm quality and concentration.

Adaptation action. Reducing the heat stress of poultry is central to effective adaptation with the introduction of more heat tolerant poutry breeds and construction of poultry housing (e.g. poultry sheds) that reduce the risk of heat stress.[16]

Poulty intervention. SHARPE has supported the establishment of more than 100 commercial poultry farms with host and refugee households, producing live birds and eggs for sale. Poultry farms established and run by refugees are small enough to be accommodated within the owners' compound inside the camp, supplying eggs to the local markets and for domestic use. The farms are linked to a local poultry firm (or poultry hub) and provided with micro-grants to co-finance start-up and initial running costs of the farms – construction of a modern poultry shed and purchase of inputs (improved breeds, feed, vaccines, medicines, equipment) from the regional poultry hub. The poultry hub provides training, advisory and technical support services and supplies inputs on an ongoing basis to the (host and refugee) poultry farmers.

For poultry farmers in the Somali region, the predicted drier and hotter conditions and increased water stress will lower egg production and reduce the weight gain of chickens produced for meat. Although the poultry sheds are well constructed and well-ventilated, with drinkers and water tanks (50 Litre) to provide water, in a future climate scenario which is both hotter and drier these sheds would not provide the cool temperatures and water needed. To become more appropriate to the changing climate, the shed design should consider how to be cooler and better ventilated; for example to include roof insulation or (solar panelled) fans, the floor would need to be built (rather than earth floor), and sheds would need to include a larger water tank and a more efficient water-delivery system to provide water throughout the day.

These climate sensitive modifications will significantly increase the costs of the sheds, for example to provide solar panels to run fans as most refugee camps are currently off-grid. Climate proofing could increase the cost of such poultry sheds by approximately 50% making this adaptation unlikely to be affordable for refugees or host communities without external grant assistance.

Sheep and Goat Trading

Climate risk. The increased frequency of weather extremes and change in water availability will have a direct effect on pasture and grazing resulting in productivity losses, and an indirect effect due to increased cost of animal housing (e.g. goat sheds) and feed and the higher prevalence of pests and disease.

Adaptation actions include improving breeds with heat and disease tolerant breeds, promoting trade and market access, providing shelter with shade and water to reduce heat stress and in flood areas with raised floors, and improving animal health through access to veterinary services and medicine.[17]

Sheep and Goat Trading intervention. Sheep and goats play an important role in household livelihoods for hosts and refugees households – it is common practice to invest in a small number of goats for production as a means for saving income (e.g. 'savings on the hoof'). For refugees, lack of access to capital keeps this activity small-scale, in most cases fewer than five animals. Building upon this common practice SHARPE supported over 40 small-scale livestock host and refugee traders to upgrade their business to a more commercial level. SHARPE also worked to build the market systems that are necessary to support sheep and goat trading including local production of nutritious feed and fodder, access to finance, access to veterinary supplies such as medicines and vitamins and peripatetic veterinary services. SHARPE micro-grants were used to construct improved sheds to house the animals and protect them from the sun; to procure animals to increase the size of the stock; and to provide training in health and nutritional requirements for fattening. Linkages were facilitated with larger livestock traders, aggregators, traders and with agro-vets.

For sheep and goat traders in Gambella, the increased severity of rainful and floods and generally wetter conditions puts the health of their livestock at particular risk as goats are vulnerable to diseases associated with wet hooves and may develop mouth infections from wet fodder (if wet fodder is cut and carried to the shed). To respond to these climate risks, infrastructure needs to protected from floods and rainfall and climate-proofed, e.g. shed floors need to be raised to protect the animals from flood waters and wet floors. Fodder storage facilities also need to be climate-proofed from floods and heavy rainfall and small solar energy generated driers for fodder should be considered. As with poultry sheds, climate proofing farm infrastructure will significantly increase the cost, potentially making it unaffordable for most refugees and poor host community households without external financing.

Agro-vets / Agri & Veterinary Inputs and Services

Climate risk. Drought and generally drier conditions pose the main climate risks to businesses offering goods and services for livestock. Due to drought, many pastoralists to move their herds to different regions, often moving to peri-urban and urban centres to better access services, or to transition to a more settled life, effectively giving up on the traditional nomadic pastoralist way of life. Drought can also cause high mortality amongst herds, reducing both livestock populations and incomes. Combined, this decreases demand for agro-vet goods and services.

Adaptation action. During extreme climate events such as the recent extended droughts, agro-vets and community animal health workers can play a critical role in supporting pastoralists to keep their animals healthy and alive. By staying open and providing their services throughout a drought they contribute to the climate resilience of households who are dependent upon livestock for their livelihoods.

Agro-vet / Agri & Veterinary intervention. To provide a sustainable supply of farm inputs, livestock supplies (vitamins, medicines etc) and veterinary services, SHARPE worked with agri-input companies and local agro-vet dealers. SHARPE offered finance to allow agro-vet dealers to invest in and expand their shops out of the main town and into refugee markets, and linked them with Addis based importers and suppliers to improve the goods and services they offer and grow their customer base in the refugee and neighbouring host communities. In Dollo Ado the SHARPE supported agro-vets are mainly refugees, operating small shops inside each of the five camps and the provide services to both hosts and refugees. Refugees have also been trained and certified as Community Animal Health Workers (CAHWs), working with the agro-vets and providing community based animal health services for a small fee.

It is important to note that agro-vets and community animal health workers are dependent on customersbeing able to afford their goods and services. For people such as refugees who are living in remote, marginal regions, climate shocks are likely to increase their need for agro-vet services. Conversely, the climate risk may reduce their ability to pay for the services and lead to a reduction on what the agro-vets and community animal health workers are able to provide, particularly to more remote areas. Therefore, during extreme climate events such as extended droughts, commercial suppliers of agricultural and vetinerary outreach services (such as agro-vets and CAHWs) should be considered an essential part of the response as ongoing support, for example cost sharing these essential supplies with these businesses, to enable them to remain functional during and beyond a crisis will contribute to overall system and household resilience.

Energy

The refugee hosting regions, and camps in both Somali and Gambella are largely off-grid[18] and therefore, most households rely on batteries and/or to buy electricity from people running diesel generators both of which are expensive and dirty, harmful to the environment and the climate.

Adaptation and mitigation actions. Solar products are overwhelmingly positive in terms of environmental and climate impact, providing much needed clean electricity to people living in off-grid situations forcing communities to use polluting and more expensive sources for their power for example, running diesel generators (or buying electricity from someone running a diesel generator) and/or batteries. In all future climate risk scenarios – hotter, drier, wetter, more unstable – the main risk to solar energy panels is from extreme weather / storms which can damage the solar panels. There may also be additional risk posed by higher temperatures but this is not a major risk and can be addressed by providing information to buyers on how to avoid damage from storms and high temperature.

Energy intervention. To provide access to clean energy SHARPE partnered with three private firms to supply and sell quality solar equipment to refugee and host community households – household solar kits; productive use kits for phone charging and/or barber shops; and solar energy kiosks. These solar kits include an after-sales service package, a two year guarantee and a PAY-GO monthly payment mechanism with reduced up-front payment for refugees. SHARPE co-funded market expansion costs of the private solar companies, with the companies importing, assembling, distributing, promoting and selling the kits while also providing after-sales services

Uba Abduselam Warsame (35) is a Somalian refugee, living in Melkadida refugee camp in the Dollo Ado region since 2019. She is a single mother of three children aged between 14 – 17.

In December 2022, SHARPE supported Uba to begin operating an energy kiosk business. She provides a mobile phone charging service, charging customers 10 ETB for a small phone and 20 ETB for a large phone. On average she charges 20 mobile phones per day. Uba also sells soft drinks and items such as shampoo, body treatments and hair oil from her kiosk. Most of her customers are refugees. She earns upto 750-800 Birr a day from her kiosk, this amount dropping to 500 Birr during times such as Ramadan or during ration cuts.

Overall the kiosk has enabled Uba to have a higher and more secure income compared to previously when she operated on a much smaller scale. The kiosk has also enabled Uba to expand and vary her income sources and therefore increase her resilience to shocks. However, solar energy products require maintenance and need both technical expertise and parts to be available and affordable. Dollo Ado's remoteness means it is a challenge to resolve issues and Uba has limited resources to afford expensive technical solutions. For solar energy products to be effective in regions such as Dollo, they need to be supplemented by local expertise to ensure the products are fully utilised. This needs to be considered (and costed) from the early stages of solar-energy interventions.

Financial Inclusion.

Digital financial services (DFS), climate risk and adaptation. Access to finance is vital to community resilience. DFS increases accessibility, enhances efficiency and speed of transactions, supports the financial inclusion of marginalized groups, such as refugees. DFS can play a vital role in building resilience by enabling vulnerable populations to prepare for severe climate-related events and recover from them.

Financial intervention. Digital financial services (DFS) are now available to refugees in Ethiopia – recent policy and bank reforms now allow refugees to own a SIM card and to open a bank account. SHARPE partnered with Shabelle Bank, to expand their DFS platform, Hello Cash, to the more remote host and refugee markets of Dollo Ado and Jijiga. SHARPE worked with Shabelle Bank to increase the number of Hello Cash agents operating in these markets to increase the number of local merchants using Hello Cash in refugee communities and to promote and onboard more than 70,000 new customers to the Hello Cash platform. Small business loans were provided to Hello Cash agents, including refugees, with personal guarantors used for collateral rather than fixed assets such as a house. Four sub-bank branches were also opened in/beside refugee camps, powered by solar energy, to increase access to financial services. Whilst there are no climate change risks to this intervention, having access to financial services can help to strengthen climate resilience of households.

Climate Risk and Adaptation Summary

Although climate adaptation and mitigation are not explicitly included in its objectives, this assessment has identified that SHARPE's interventions are making significant contributions in strengthening the climate adaptation and resilience of refugees and host communities, and is lowering GHG emissions (mitigation). Climate adaptation is strengthened by increasing income, assets, and diversifying livelihoods, which together increase the ability of individuals and households to adapt to climate change stresses, shocks and variability and/or help reduce exposure to them. This is being achieved by:

- Creating new opportunities for income that are well suited for refugees – poultry, goat and energy interventions are purposely designed for small compounds and for women's participation Upgrading infrastructure used in these agri-businesses (i.e. sheds) already provides a higher level of protection for the animals against climate extremes than would otherwise be the case.
- Improving access to digital financial services and business financing increases financial inclusion and contributes to adaptation and resilience. Digital financial services contributes to adaptation by increasing accessibility, enhancing efficiency and the speed of transaction, and enabling the financial inclusion of refugees. It provides income and supports the development, distribution or adoption of new technologies that will enable access to finance. Business financing for agri-business plays a significant role in their success and are essential for providing the resources that enable small businesses to successfully operate and expand which in turn enables their role in provision of services and resources for household / community adaptations.
- New 'off-grid' solar systems allows the delivery of renewable energy directly to houses, businesses and/or community services and diminishing use of diesel generators – contributing to both adaptation and mitigation.

Diversification with multiple sources of income is central to adaptation, enabling households to spread risk and being able to piece together a livelihood from different income streams. Even if one activity brings in more income, other activities are often initiated or pursued. Diversification is a traditional adaptation tothe high dependency on rainfall patterns as demonstrated in the Somali region pastoralist, agro-pastoralist and crop producing communities (see Box 3 below) as households today combine livestock and crop farming with remittances and gifts, casual labour, own business(es), salaried work loans or humanitarian aid.[19]

Case study – Diversification > Adaptation

Abubakar Hassen, is a 45-year-old Somali refugee living in Buramino Camp, Dollo Ado. He arrived in Ethiopia in 2009 when he was forced to leave due to conflict and drought in his homeland. Before migrating, Abubakar owned multiple business ventures including in agriculture, livestock and retail.

As a refugee entrepreneur Abubakar manages several businesses which provide him with different income sources. In 2022, SHARPE supported Abubakar with a 250,000 ETB grant (£3,900) for shed construction and purchase of 20 goats. Through this support, he has increased the number of goats he can sell in each 3-month cycle from 20 to 70 goats and this additional income has enabled him to expand and diversify his other income generating activities.

In addition to trading goats, Abubakar operates a small self-financed mill (to process grains, primarily food rations), farms vegetables, trades charcoal, owns a retail shop and also owns livestock (cattle and camels) being cared for by clan/kin in another area. This gives him an estimated monthly income of 24,000 ETB (approximately £300). While he does not own land to farm, he rents land from a local landowner and grows onions with three other investors. In 2023 his onion crop was lost due to flooding and Abubakar suffered financial losses of around 1million ETB (£15,600) but he plans to invest in onions the next season.

Challenges such as the decrease in rations during 2023 which impacted his feed mill business and flooding which caused financial loss in his farming business continue to affect Abubaker. SHARPE's support has enabled him to strengthen his businesses and improved his ability to recover from shocks and his multiple income streams have given him the adaptive capacity to respond to the changing conditions.

However, the findings also suggest that if climate change had been integrated into the SHARPE programme from the beginning the results would be greater. There were missed opportunities to contribute to climate adaptation and mitigation. As the climate becomes drier, hotter, wetter and more unpredictable (depending upon the region) agri-business and farm based livelihoods are at greatest risk from climate change. Since this assessment was conducted in the arid conditions of Dollo Ado it focused more on the health of small farm animals, which is where SHARPE has focused in this region. With small modifications, for example climate proofing small infrastructure such as poultry and goat sheds, these contributions can be improved, however the additional costs required to climate proof farm infrastructure must be considered. Incorporating additional financing in the intervention design and budget to make these sheds climate proof and provide solar power can further lower climate risk and contribute to the sustainable adaptation of the interventions to the changing climate.

It is crucial for vulnerable refugee communities, not only in Ethiopia but in other countries and regions at high climate risk, that climate resilience moves to the centre of programme design, planning and implementation including costing. Over three-quarters of refugees globally (15.7 million in 2020) are in a protracted situation and new approaches are needed to drive the transition to sustainable development AND sustainable adaptation.[20] Anticipating climate-related risks holistically is key in steering humanitarian responses away from away from reactive towards proactive and resilient developmental paths.[21]

A significant barrier to programmes increasing their climate contribution is that climate change and its related risks are not effectively considered and integrated into programme design and planning. Climate risk is not considered as a primary driver for shaping economic development strategies, often only briefly mentioned in background statements and actions for adaptation are 'add-ons' that are not adequately budgeted. As a result opportunities continue to be missed for strengthening adaptation and lowering emissions (e.g. integration of clean energy into programme targets). To address this, a key step would be the use of a climate lens during programme design that would put to the forefront the risks, vulnerabilities and impacts related to climate variability and change. The programme context and actions should demonstrate a clear and direct link between the identified risks, vulnerabilities and impacts and the specific project activities. By using a climate lens during programme design, for example, business and services can be recognised as contributing to climate change adaptation in a number of ways e.g. by diversifying income opportunities for communities that cannot continue their traditional way of life as a result of climate change (such as is occurring in areas experiencing increasing climate risk), but also by supporting the development, distribution or adoption of new technologies to better deal with climate change.

UK International Climate Finance (ICF): Contribution to adaptation and mitigation

International Climate Finance (ICF) is Official Development Assistance (ODA) from the UK allocated to support developing countries to reduce poverty and respond to and mitigate the challenges caused by climate change and environmental degradation. While SHARPE is not funded by ICF, this study has considered whether SHARPE's results meet the criteria. ICF uses a Key Performance Indicator (KPI) framework for reporting which includes 17 KPIs. Two of these KPIs are of direct relevance to SHARPE - ICF KPI 1 and ICF KPI 2.1.

UK ICF KPI 1 – Adaptation

Number of people supported to better adapt to the effects of climate change (as a result of ICF). This output indicator counts the number of people (total direct and total indirect beneficiaries) whom ICF programmes have supported to prepare and deal with the effects of climate change, including long-term changes in weather patterns and the increasing frequency and severity of extreme weather events.

KPI 1 is an output indicator that measures the reach of UK ICF climate change adaptation programmes. It counts the total number of people supported by ICF to be better able to adapt to the effects of climate change but does not measure whether this support was successful in reducing people's vulnerability to effects of climate change To meet the KPI criteriaa, the programme is expected to reduce people's vulnerability to the effects of climate change by giving them **resources, tools, or skills to better cope with changing weather patterns**, including more unpredictable and extreme weather, as well as long-term, more incremental changes.

As of October 2024, SHARPE reported **194,045 people** were more resilient and had benefitted with improved business and/or household income through either increased income, cost savings, a new or better job and/or increase in asset value. This counts 'direct' beneficiaries – those that have worked directly with SHARPE to establish, adapt, innovate or expand a business model – and 'indirect' beneficiaries – those that benefit from accessing improved goods or services from SHARPE supported businesses, such as agro-vet customers or DFS users. Deducting people benefitting from improved access to clean energy (reported below for KPI 2.1) this assessment concludes that **181,778 people** benefitting with an improved income, cost savings or a new/ better job by having a stronger and more resilient business / livelihood, are contributing to KP1.

UK ICF KPI 2.1 - Clean energy (Mitigation)

Number of people and social institutions with improved access to clean energy (as a result of ICF). This is an output indicator that counts the number of people who have improved access to clean energy as a result of an ICF intervention. UK ICF KP2.1 criteria includes interventions such as the installation of solar technology; green mini-grids for local power supply in rural areas; clean cook stoves that reduce people's dependence on firewood; and interventions that support improvements in energy efficiency.

Through SHARPE's work on solar energy, as of October 2024, **14,266 people** had improved access to clean energy, either through the purchase of a household solar kit or through access to enery services supplied by an energy kiosk.

Key Lessons

1. Refugees are often housed in marginal areas that are climatic hotspots. Gambella and Somali regions in Ethiopia host more than 700,000 refugees and are predicted to become hotter, drier, wetter with more frequent extreme climate events and greater unpredictability.
2. Market systems development programmes such as SHARPE, operating in fragile settings, have the potential to make significant contributions to resilience, climate adaptation and mitigation and contribute to ICF results. However, care should be taken to ensure that interventions are climate proofed and remain fit-for-purpose in future climate scenarios.
3. Climate proofing investments, in this case sheds built to house poultry and goats, requires additional funding. Interventions should integrate climate considerations in their design along with further investments in solar energy and access to financial services, all of which requires additional financial support.
4. Climate change considerations need to be integrated from the beginning of programme planning for all climate-sensitive sectors, particularly those seeking to improve the livelihoods of people living in crisis in climate vulnerable, marginal regions.

Conclusions

Ethiopia is one of most vulnerable countries in the world to climate change. Within Ethiopia, refugee settlements are in marginal regions that are climatic 'hotspots,' intensified in some areas by refugee-host community conflict, conflict which is often driven by competition over natural resources, in turn exacerbated by climate change. With more than two-thirds of the refugee population housed in the Somali and Gambella regions this is the situation in Ethiopia. Under climate change scenarios, these regions are predicted to become hotter, drier, wetter, and with more frequent extreme climate events and greater unpredictability.

Climate change is a threat multiplier for refugee and host communities with the potential to undermine climate-sensitive livelihoods and destroying assets and resources of people already living in crisis. Given the protracted nature of many refugee settings, climate change considerations need to be integrated into any intervention that seeks to sustainably improve livelihoods and self-reliance. Considering climate risk in investments aiming at fostering self-reliance of long-term refugees becomes even more pivotal as it holds the potential to reduce the critical need for on-going assistance over the mid-to long-term.

Programmes such as SHARPE can provide evidence that a more systemic market systems development approach can succeed and be a model for sustainable development and sustainable adaptation. This assessment found that even though climate risk was not considered and integrated into its initial planning, SHARPE's interventions were making significant contributions to climate adaptation and mitigation among refugees and remote host communities and are contributing to the UK International Climate Finance results under KPI 1 (adaptation) and KPI 2.1 (clean energy). The assessment also found that SHARPE could do even better by adapting some of its interventions, particularly investments in farm level infrastructure constructed for poultry and livestock to ensure that this infrastructure can serve its purpose in future climate scenarios.

Looking to the future, the assessment findings highlight the importance of including climate risk from the very beginning in the planning and design of programmes for protracted refugee communities to better enable the long-term sustainability of initiatives in hotter, drier yet flood prone regions. This is particularly the case for programmes seeking to improve the livelihoods, self-reliance and economic inclusion of refugees living in climate vulnerable, marginal regions. To be sustainable these programmes have to promote climate resilient livelihoods and support climate-sensitive economic inclusion.

Endnotes

1 The widely used ND Gain Index summarizes a country's vulnerability to climate change and other global challenges in combination with readiness to improve resilience. https://gain.nd.edu/our-work/country-index/rankings/.

2 Vulnerability measures the exposure, sensitivity, and ability to cope with climate related hazards by accounting for the overall status of food, water, environment, health, and infrastructure within a country. Readiness targets those portions of the economy, governance and society that affect the speed and efficiency of adaptation.

3 Edgar (Emissions Database for Global Atmospheric Research] (2023) https://edgar.jrc.ec.europa.eu/report_2023#:~:text=China%2C%20the%20United%20States%2C%20India,61.6%25%20of%20global%20GHG%20emissions.

4 BMZ, PIK, GIZ and KfW (German Federal Ministry for Economic Cooperation and Development, Potsdam Institute for Climate Impact Research, Deutsche Gesellschaft für Internationale Zusammenarbeit, and Kreditanstalt für Wiederaufbau). n.d. Climate Risk Profile: Ethiopia. https://www.pik-potsdam.de/en/institute/departments/climate-resilience/projects/project-pages/agrica/climate-risk-profile_ethiopia_en

5 USAID (2024) Climate Change Risk Profile Ethiopia. https://www.climatelinks.org/resources/climate-risk-profile-ethiopia

6 World Bank Group (2020) Climate Risk Country Profile: Ethiopia. https://climateknowledgeportal.worldbank.org/sites/default/files/2021-05/15463A-WB_Ethiopia%20Country%20Profile-WEB.pdf

7 https://data.unhcr.org/en/country/eth

8 Dire Dawa, Gambella, Somali, Oromia and the former SNNPR regions are the most vulnerable to climate change.

9 The research did not include the refugee communities; its focus was on the host community and IDPs.

10 Reach (March 2024) The Impact of Drought and Climate-Related Shocks on Livelihood Practices in Somali (Factsheet) https://repository.impact-initiatives.org/document/reach/0579137a/REACH_ETH_Factsheet_SomaliLivelihoods_1504.pdf

11 https://thewaterchannel.tv/videos/prosopis-in-somaliland-a-tree-as-threat-and-opportunity/

12 Getahun et al. (2021) Determinants of climate variability adaptation strategies: A case of Itang Special District, Gambella Region, Ethiopia. Climate Services, Vol.23, August 2021. https://www.sciencedirect.com/science/article/pii/S2405880721000339#:~:text=The%20Gambella%20Peoples%20National%20Regional,over%2Dflow%20of%20major%20rivers

13 https://www.ipcc.ch/site/assets/uploads/2018/02/WGIIAR5-Chap20_FINAL.pdf

14 https://www.sipri.org/sites/default/files/2024-04/pb_2404_mistra_livelihood_diversification_v2.pdf

15 Recent research suggested that pastoral households participating in non-farm activities in addition to livestock rearing were at a better economic level than those who did not engage in the same activities. https://www.tandfonline.com/doi/full/10.1080/23311886.2023.2277349#d1e184

16 https://www.thepoultrysite.com/articles/climate-change-and-poultry-production

17 https://agrica.de/wp-content/uploads/2020/09/Climate-Risk-Analysis_Ethiopia_EN-1.pdf

18 Ethiopia has the third largest energy access deficit in Sub-Saharan Africa with about half the population still without access to reliable electricity. The grid network covers nearly 60% of town and villages, but only 10% of Somali is on the national grid. Ethiopian electricity production heavily relies on hydropower, which is vulnerable to climate related changes to hydrological conditions, stream flow, water availability, and reservoir storage. https://www.worldbank.org/en/news/press-release/2024/04/03/energizing-afe-ethiopia-new-world-bank-program-expands-access-to-electricity#:~:text=Ethiopia%20has%20the%20third%20largest,without%20access%20to%20reliable%20electricity.

19 Reach (March 2024) The Impact of Drought and Climate-Related Shocks on Livelihood Practices in Somali (Factsheet) https://repository.impact-initiatives.org/document/reach/0579137a/REACH_ETH_Factsheet_SomaliLivelihoods_1504.pdf

20 https://www.unhcr.org/neu/wp-content/uploads/sites/15/2021/08/10-facts-about-refugees-2021-ENG_Vi2.pdf

21 UNHCR (2023) East and Horn of Africa and Great Lakes Region. Climate Action Plan (2023-2028) https://data.unhcr.org/en/documents/details/104399

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